

Paid Acquisition & Revenue Analytics

Project brief: business context, rules, data and analysis plan

Project overview

Bolo is an Android app that helps Indian users practise spoken English. A new user pays ₹2 to start a 24-hour trial. When the trial ends, a paid plan starts automatically: either ₹249 a month or ₹1,499 a year. Every user is bought through paid advertising, across three ad channels (josh, inmobi and glance) and six campaigns.

Bolo is a fictional company. Its campaigns and data were created for this project, and every rule below is treated as true.

Business problem

Bolo's growth team funds its six campaigns roughly in line with what each one spent over the past three months. That split was never tested against what really matters: how many paying users each campaign brings. The August budget has now been set at ₹1 crore, about 17% above the recent monthly spend, and two concerns sit on top of it:

- **The co-founder** has noticed that some campaigns produce very cheap trials, and isn't sure cheap trials turn into cheap customers.
- **Finance** has warned that the budget could be cut to ₹60 lakh.

Main business question. Where should August's ₹1 crore of ad budget go, and what changes if it is cut to ₹60 lakh?

The analysis has to end in a budget plan: spend per campaign, with the trials, paying users and first-month revenue that spend is expected to buy.

Business rules (treated as true)

Trials and paid plans

- A ₹2 purchase starts a 24-hour trial. When the trial ends, the paid plan is charged automatically: this charge is a renewal.
- So the renewals on any day belong to the trials started on the day before.
- Each trial is assigned to the monthly plan (₹249) or the annual plan (₹1,499) when it starts. `annual_renewals` counts the annual ones.

Revenue and refunds

- Gross revenue on a day = trials × ₹2 + monthly renewals × ₹249 + annual renewals × ₹1,499.
- Refunds and chargebacks are booked on the day they happen, which can be later than the original charge.
- All money is in rupees and excludes GST.

Acquisition

- Every install, trial and renewal is attributed to one campaign. Spend is what the ad channel charged that day.
- The data covers 1 May to 29 July 2026: 90 days.

The data

`bolo_growth_data.csv`: 540 rows, one row per campaign per day (6 campaigns × 90 days).

Column	Meaning
<code>date</code>	Day of activity (IST)
<code>channel, campaign</code>	Ad channel (josh, inmobi or glance) and campaign name
<code>spend_inr</code>	Ad spend that day
<code>impressions, clicks, installs</code>	The ad funnel, attributed to the campaign

trials	₹2 trials started that day
renewals	Paid plans charged that day (24 hours after the trial)
annual_renewals	The part of renewals on the ₹1,499 annual plan
revenue_inr	Gross revenue booked that day, trials included
refunds_inr	Refunds and chargebacks booked that day

Analysis plan

The analysis is organised in seven parts, each answering one piece of the budget decision.

Part 1 • Data validation

- Profile the file: completeness, ranges, duplicates, impossible values.
- Test every rule the data must follow: the funnel only narrows (impressions > clicks > installs > trials > renewals), renewals never exceed the previous day's trials, annual renewals never exceed renewals, revenue matches prices × plans, and cost per thousand impressions stays plausible.
- Record each problem in an investigation log: what was noticed, rows and ₹ affected, what probably happened, and how it was handled (corrected, excluded, flagged but kept, or left alone), with the reason. Spend and revenue are not changed without proof.
- List the questions an engineer would need to answer before real money depends on the numbers.

Part 2 • Campaign performance

One row per campaign for the full period: cost per install, cost per trial, trial-to-paid rate, cost per paying user, annual plan share, and first-month return on ad spend (ROAS) after refunds.

Key pitfall: renewal timing. Dividing a day's renewals by the same day's trials compares two different groups of people. When a campaign's spend is rising or falling, that ratio is biased in a predictable direction, and averaging daily ratios adds a second bias. Each day's trials must be matched to the next day's renewals.

Part 3 • Cheap trials vs cheap customers

- Rank the campaigns by cost per trial, then by cost per paying user. Does the order change?
- Explain, without any query or formula, whether the cheapest-looking campaigns are good value or a trap.

Part 4 • Weekdays and weekends

Compare cost per paying user on weekdays and weekends for each campaign, by the day the trial started. Should spend be scheduled differently?

Part 5 • Budget plan

- ₹1 crore plan: one row per campaign with proposed spend, expected trials, expected paying users and expected first-month revenue.
- Diminishing returns: test whether each extra rupee buys less on high-spend days, and decide whether the plan should assume it.
- Assumptions listed in full, including how far the plan trusts spend levels never tested before.
- The campaign to cut, and the evidence that would prove that choice wrong.
- ₹60 lakh scenario: which campaigns to protect first.
- One headline number for the whole plan, and why that one.

Part 6 • One chart

One chart, not a dashboard, that the co-founder could read in ten seconds and know what to do, with one sentence underneath saying what it tells them to do.

Part 7 · Weekly steering metric

Choose the single metric the team should steer on every week, and argue for it in three sentences against the alternatives (cost per trial, ROAS).

Approach

- Open questions are written down, each with the assumption used meanwhile, instead of guessing silently.
- No machine learning. Every question here can be answered with careful cohort matching, grouping and a simple regression line, so a model would add complexity without adding insight.
- Every recommendation has a rupee amount or a user count attached.

Deliverables

- Analysis notebook: a Python notebook that reruns top to bottom from the CSV.
- Budget report: recommendation first, evidence second, caveats last.
- Charts, including the one chart for the co-founder.

Quality standards

Area	What good looks like	What to avoid
Data judgement	Each problem found from its symptom, handled deliberately, with the ₹ attached	Deleting or interpolating rows without saying so
Grain	Trials matched to next-day renewals; flagged days kept out of ratios	Same-day ratios; averages of daily ratios
Metrics	Cost per paying user, not cost per trial	Ranking campaigns on the cheapest-looking number
Planning	Diminishing returns tested and stated; no plan outside tested spend	Putting all the money into the best campaign
Decision	One plan, one campaign to cut, and what would change the decision	A table of options with no choice
Communication	The co-founder understands the report without opening the notebook	The notebook is the answer